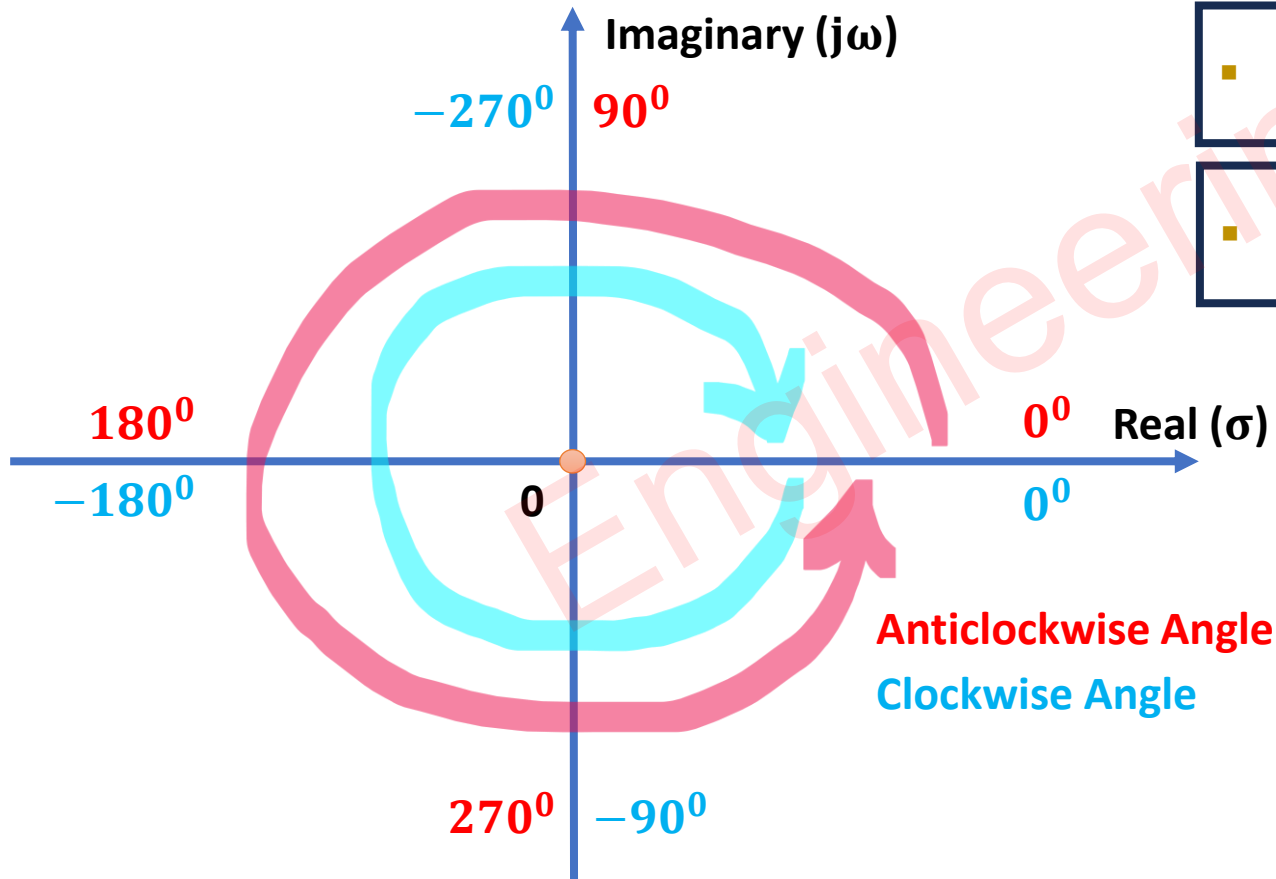


Polar Plot

Basics of Polar Plot

- ❑ Polar Plot is used for frequency Response Characteristics of the System.
- ❑ Polar Plot is a plot of Magnitude and Phase by varying frequency from Zero to infinite.
- ❑ Polar Plot is plotted on a Real Imaginary Plane with a Circular Pattern.



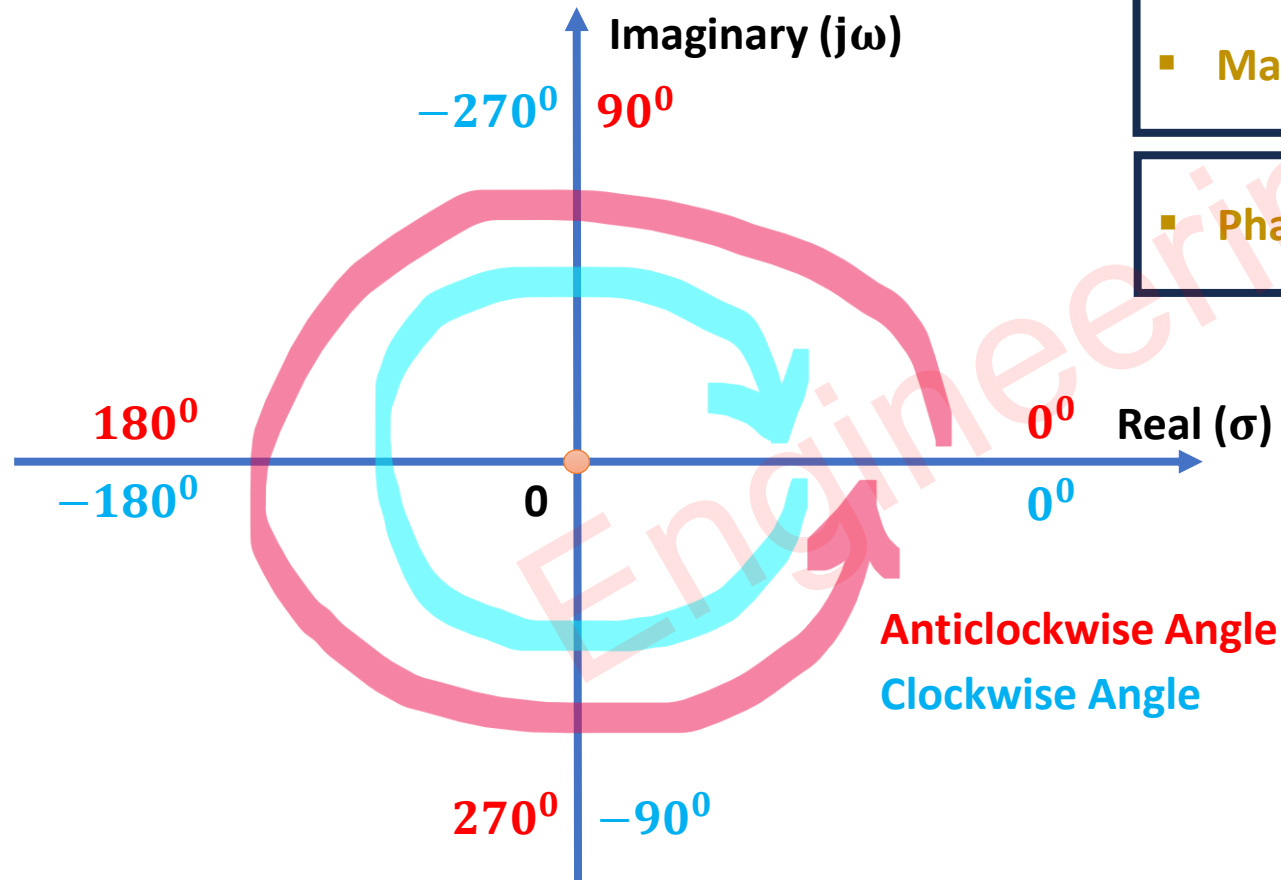
▪ Magnitude = Distance with respect to Center.

▪ Phase = Angle with respect to Real Axis.

Procedure of Polar Plot

- ❑ Polar Plot is a plot of Magnitude and phase by varying frequencies from Zero to Infinity.
- ❑ Polar Plot is Plotted using Open Loop Transfer Function $G(S)$
- ❑ Put $S = j\omega$,

$$\Rightarrow G(j\omega) = |G(j\omega)| \angle G(j\omega)$$



▪ Magnitude = Distance with respect to Center.

▪ Phase = Angle with respect to Real Axis.

Procedure of Polar Plot

❑ Step – 1: Determine Open Loop Transfer Function $G(S)$.

❑ Step – 2: Put $S = j\omega$,

$$\Rightarrow G(j\omega) = |G(j\omega)| \angle G(j\omega)$$

❑ Step – 3: Find Magnitude $|G(j\omega)|$ and Phase $\angle G(j\omega)$, at $\omega = 0$ and $\omega = \infty$.

❑ Step – 4: Separate Real and Imaginary Part of $G(j\omega)$. It can be done by Multiplying and Dividing Complex Conjugate of denominator.

$$\Rightarrow G(j\omega) = \text{Real} [G(j\omega)] + j \text{ Imag} [G(j\omega)]$$

❑ Step – 5: If $\text{Real} [G(j\omega)] = 0$, Then we will get Intersection to Imaginary axis.

❑ If $\text{Imag} [G(j\omega)] = 0$, Then we will get Intersection to Real axis.

Advantages of Polar Plot

- ❑ Single Plot plots Frequency Response Characteristics {Magnitude and Phase}.
- ❑ Graphical Study of Stability is Easy with Polar Plot {Compared to Root Locus and Bode Plot}.
- ❑ Easy to determine Gain Cross Over Frequency ω_{GC} and Phase Cross Over Frequency ω_{PC} using Polar Plot.
- ❑ Polar Plot can be plotted from the Open Loop Transfer Function {Other techniques need Closed Loop Transfer Function}.

Polar Plot Parameters

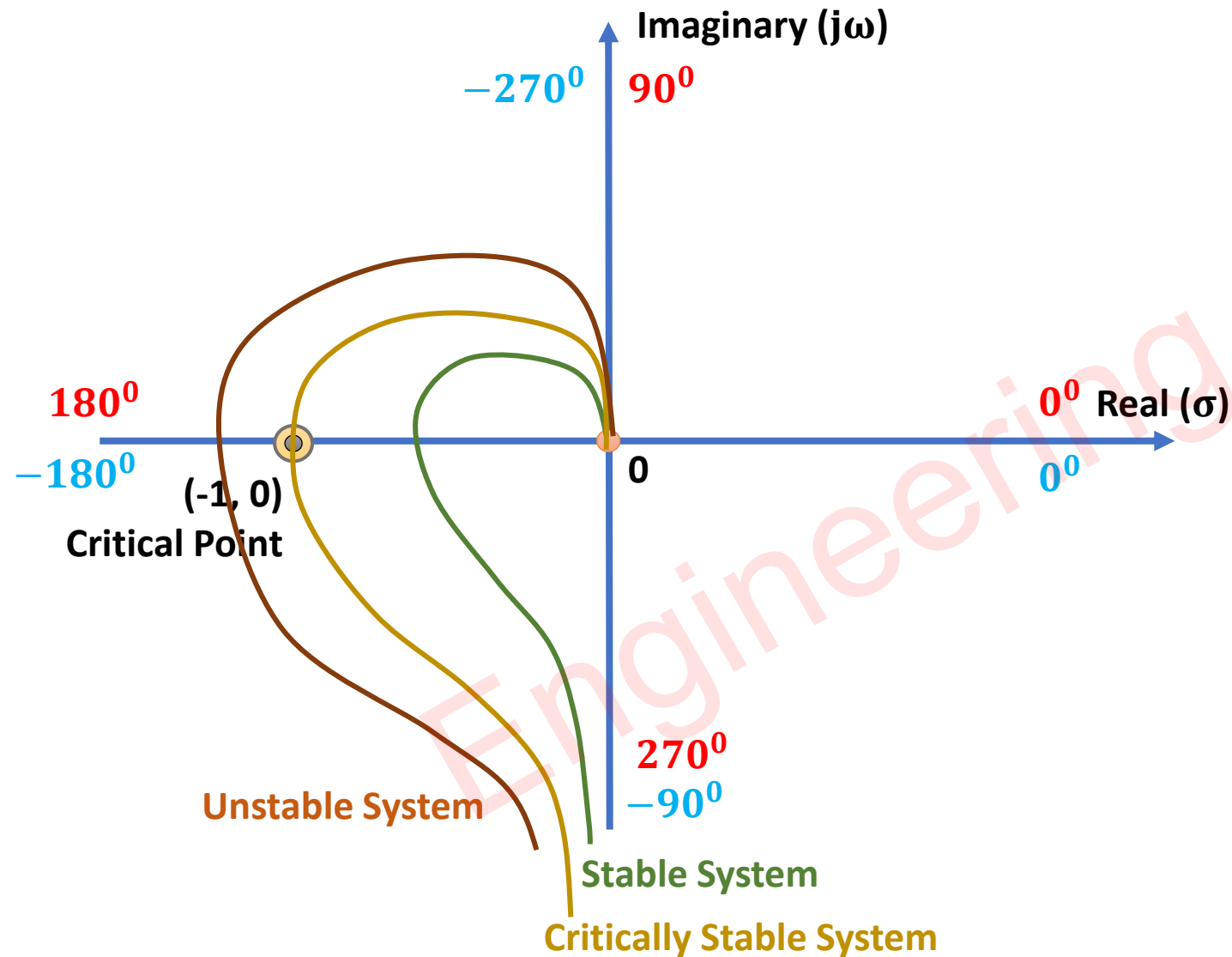
Outlines

- ❑ Critical Point and Stability using Polar Plot
- ❑ Phase Cross Over Frequency
- ❑ Gain Margin
- ❑ Phase Margin
- ❑ Gain Cross Over Frequency
- ❑ Stability Analysis using Polar Plot

Engineering Funda

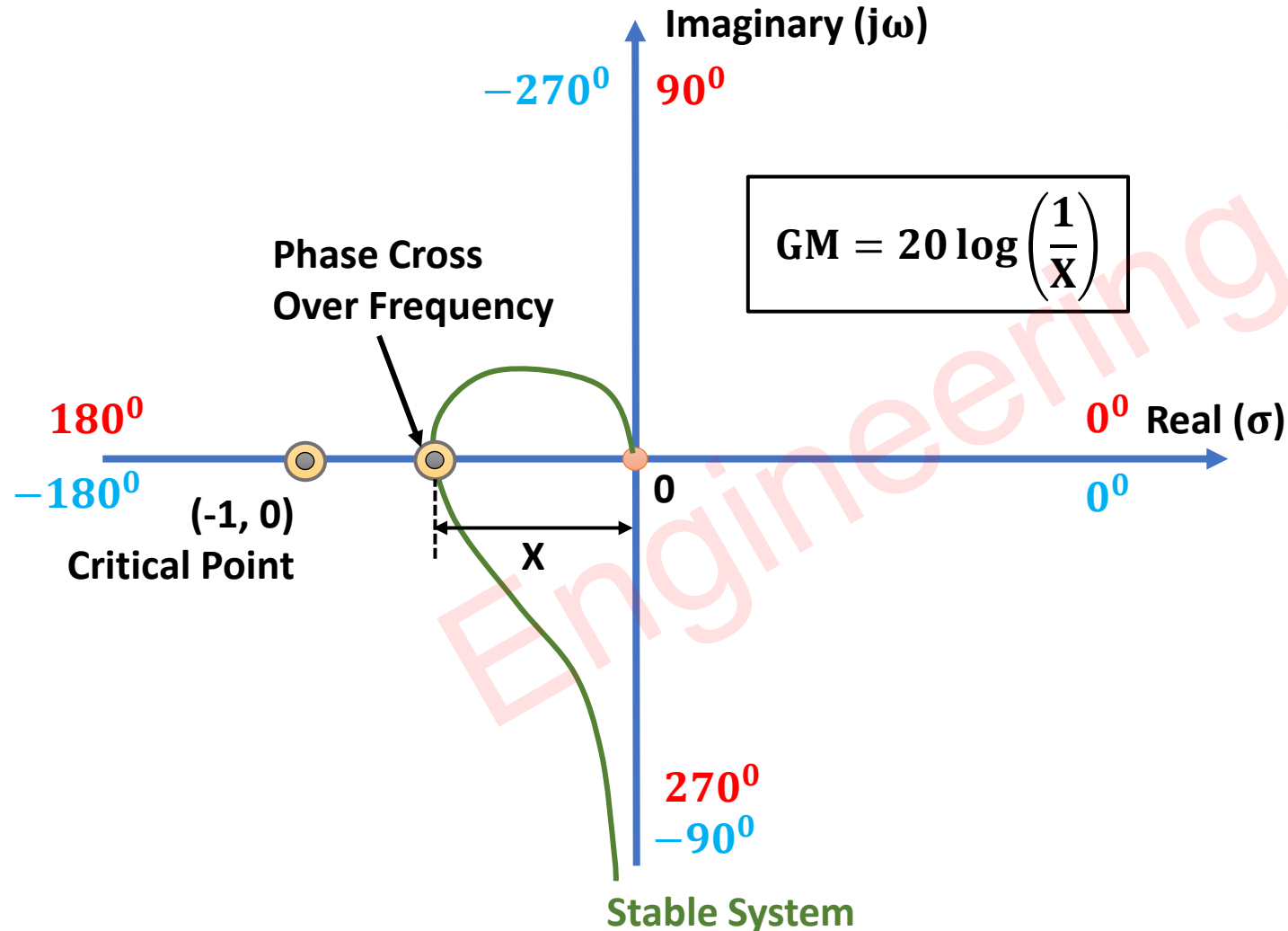
Critical Point and Stability of Polar Plot

□ $(-1,0)$ Point is Critical Point in Polar Plot.



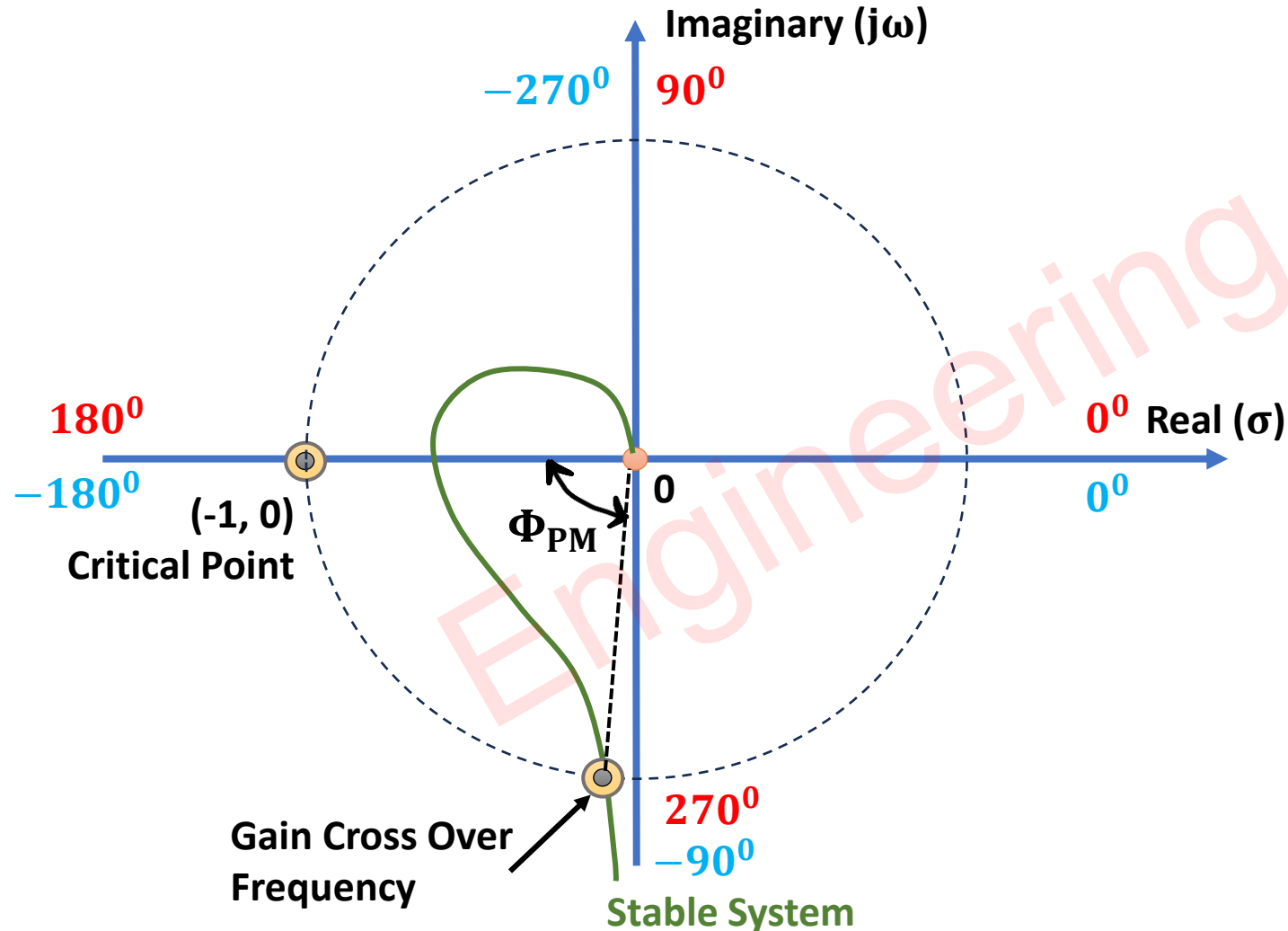
Phase Cross-Over Frequency and Gain Margin

- ❑ Phase Cross-Over Frequency: It is a frequency at which Phase is crossing -180° Angle.
- ❑ Let's Assume, Magnitude at Phase Cross-Over Frequency is X.
- ❑ Gain Margin: It is a Margin with respect to Stable System.



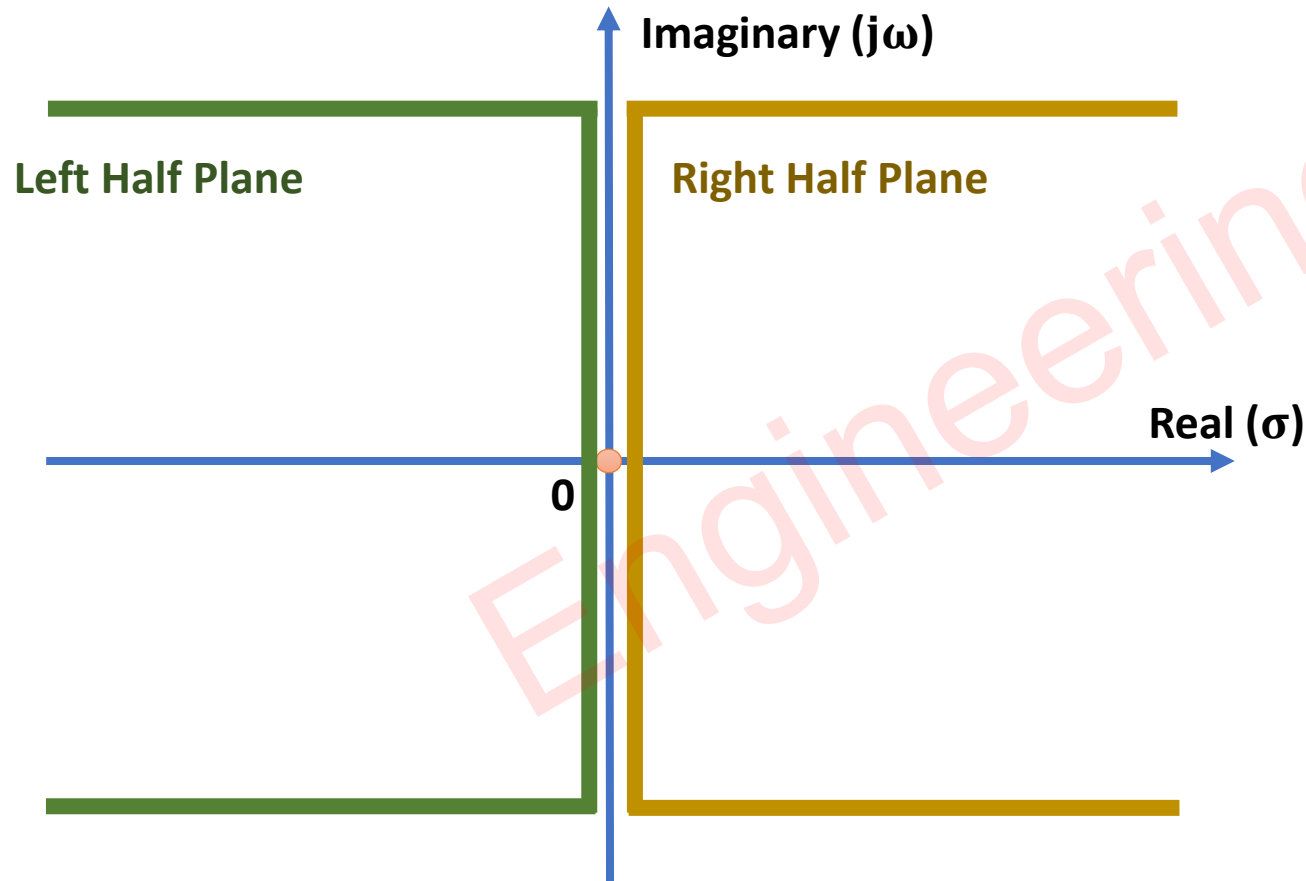
Gain Cross-Over Frequency and Phase Margin

- **Gain Cross-Over Frequency:** It is a frequency at which Gain is Crossing Unit Magnitude.
- **Phase Margin:** It is a Angle Measured from -180^0 in anticlockwise direction to the point of Gain Cross-Over Frequency.



Stability Analysis using Polar Plot

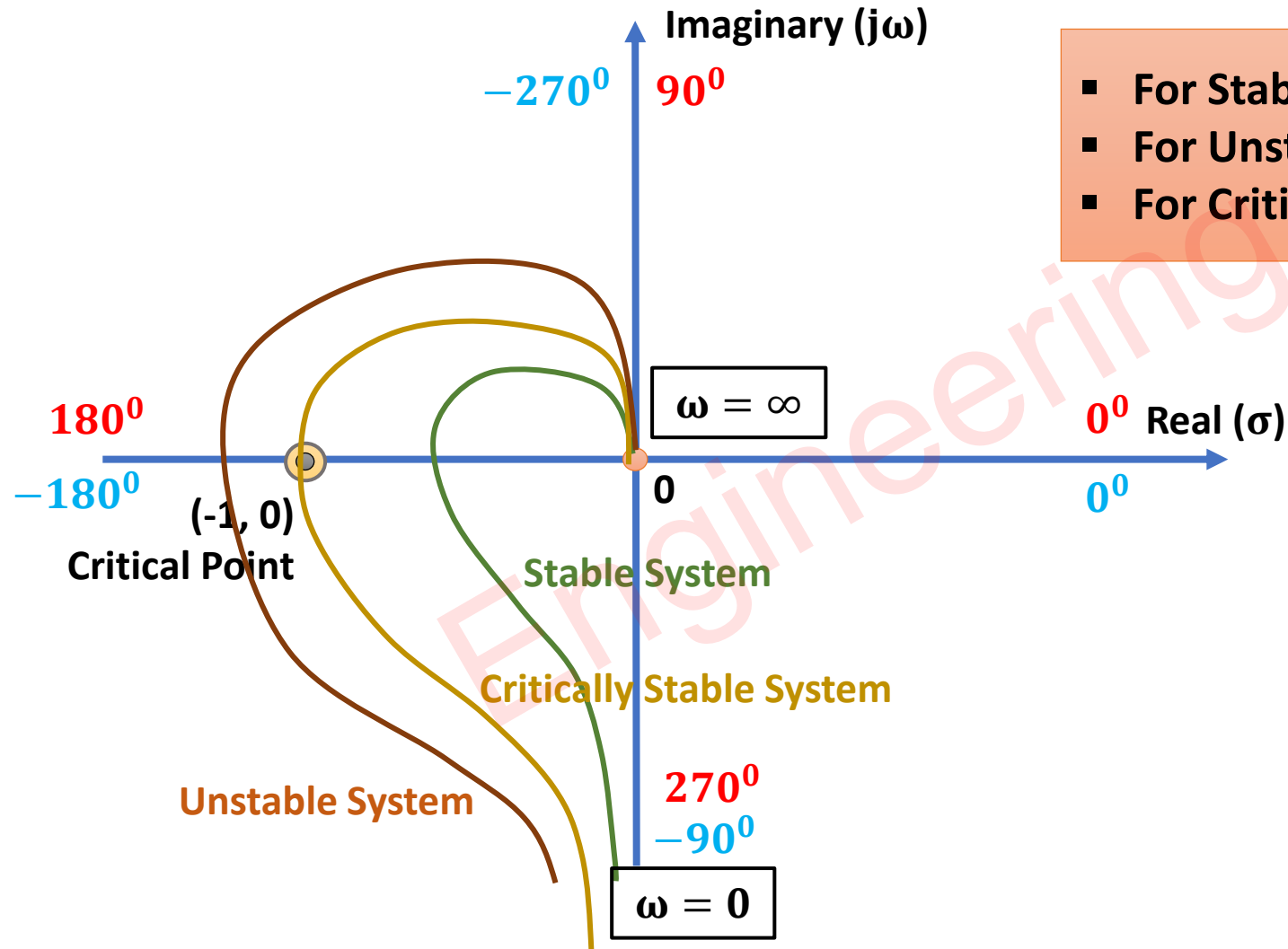
- ❑ Polar Plot Stability Analysis is only applicable to minimum phase system.
- ❑ For Minimum Phase System, all the Poles and Zeros are there in Left Half Plane of the S-Plane. Polar Plot is possible for all the systems but for stability analysis all the Poles and Zeros should be there in Left Half Plane of S-Plane.



Stability Analysis using Polar Plot

❑ To check Stability, Check Polar Plot around Critical Point $(-1, 0)$.

- If $(-1, 0)$ is enclosed by Polar Plot {System is Unstable}.
- If $(-1, 0)$ is not enclosed by Polar Plot {System is Stable}.

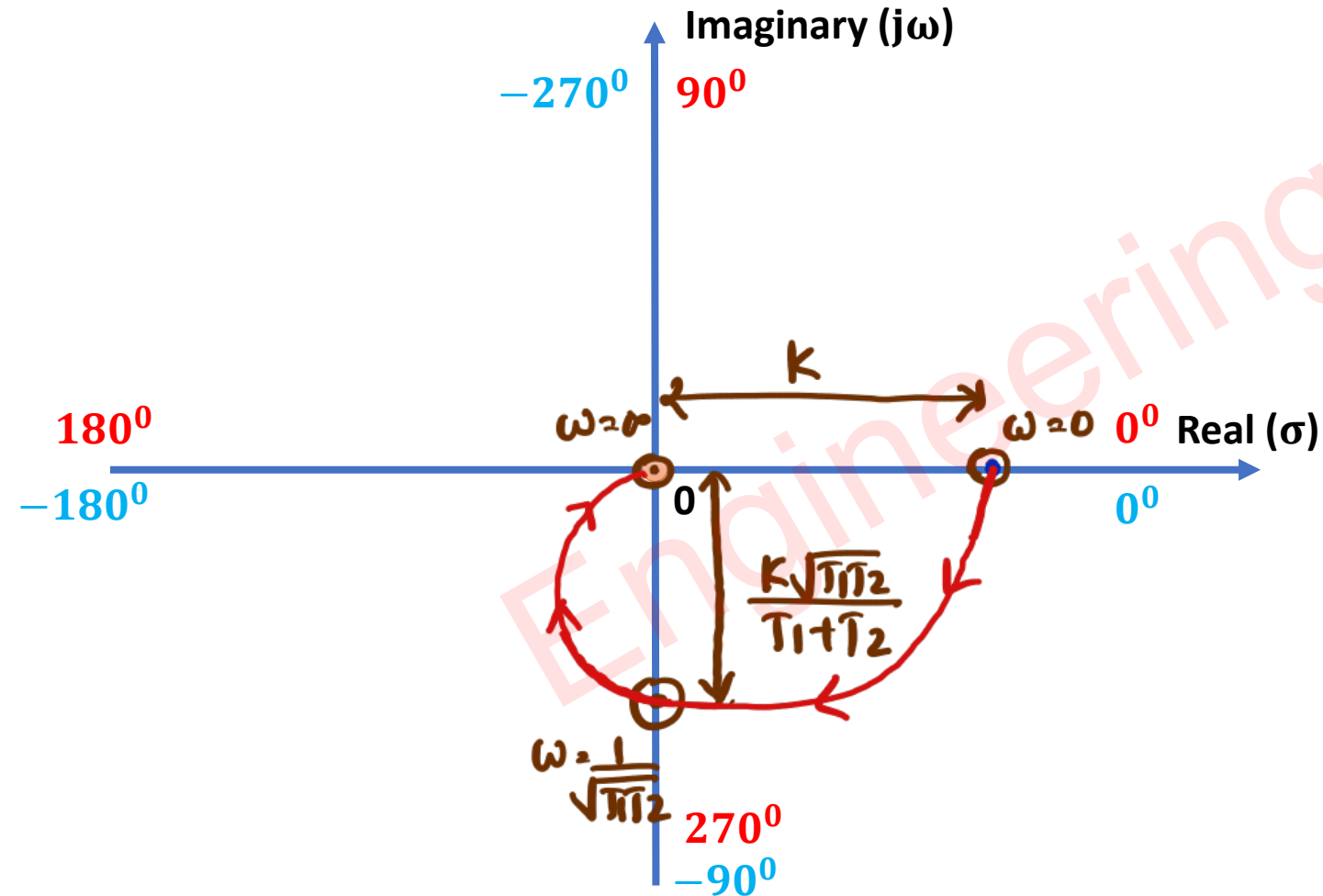


- For Stable System, $\omega_{PC} > \omega_{GC}$ and $GM = +Ve$
- For Unstable System, $\omega_{PC} < \omega_{GC}$ and $GM = -Ve$
- For Critically Stable System, $\omega_{PC} = \omega_{GC}$ and $GM = 0$

Polar Plot For Type '0' System

Example 1: Draw Polar Plot for given Open Loop Transfer Function

$$\Rightarrow G(S) = \frac{K}{(1 + ST_1)(1 + ST_2)}$$



Step-1 - Put $s = j\omega$ in $G(s)$

$$\Rightarrow G(j\omega) = \frac{K}{(1+j\omega T_1)(1+j\omega T_2)}$$

Step-2 - Write $G(j\omega)$ in polar Form

$$\rightarrow G(j\omega) = |G(j\omega)| \angle G(j\omega)$$

$$\Rightarrow \text{magnitude} = |G(j\omega)| = \frac{K}{\sqrt{1+\omega^2 T_1^2} \sqrt{1+\omega^2 T_2^2}}$$

$$\begin{aligned} \Rightarrow \text{phase} &= \angle G(j\omega) \\ &= -\tan^{-1}(\omega T_1) - \tan^{-1}(\omega T_2) \end{aligned}$$

Step-3 - Find magnitude and phase at $\omega = 0$ & $\omega = \infty$

- At $\omega = 0$, $\text{mag} = K$, $\text{phase} = 0$

- At $\omega = \infty$, $\text{Mag} = 0$, $\text{phase} = -180$

Step 4 - Separate Real & Imag. parts of $G(j\omega)$

$$\Rightarrow G(j\omega) = \frac{k}{(1+j\omega T_1)(1+j\omega T_2)} \times \frac{(1-j\omega T_1)(1-j\omega T_2)}{(1-j\omega T_1)(1-j\omega T_2)}$$

$$\Rightarrow G(j\omega) = \frac{k(1-\omega^2 T_1 T_2) - k j (\omega T_1 + \omega T_2)}{(1+\omega^2 T_1^2)(1+\omega^2 T_2^2)}$$

$$\Rightarrow G(j\omega) = \underbrace{\frac{k(1-\omega^2 T_1 T_2)}{(1+\omega^2 T_1^2)(1+\omega^2 T_2^2)}}_{\text{Real } [G(j\omega)]} - j \underbrace{\frac{k(\omega T_1 + \omega T_2)}{(1+\omega^2 T_1^2)(1+\omega^2 T_2^2)}}_{\text{Imag } [G(j\omega)]}$$

Step 5 - Intersection to Real Axis $\Rightarrow \text{Imag } [G(j\omega)] = 0$

$$\Rightarrow \omega = 0$$

Step 6 - Intersection to Imag Axis $\Rightarrow \text{Real } [G(j\omega)] = 0$

$$\Rightarrow 1 - \omega^2 T_1 T_2 = 0$$

$$\Rightarrow \omega = \frac{1}{\sqrt{T_1 T_2}}$$

$$\Rightarrow \text{magnitude} = |G(j\omega)| = \frac{k}{\sqrt{1 + \omega^2 T_1^2} \sqrt{1 + \omega^2 T_2^2}}$$

$$= \frac{k}{\sqrt{1 + \frac{T_1}{T_2}} \sqrt{1 + \frac{T_2}{T_1}}}$$

$$= \frac{k \sqrt{T_1 T_2}}{T_1 + T_2}$$

$$\Rightarrow \text{phase} = \angle G(j\omega)$$

$$= -\tan^{-1}(\omega T_1) - \tan^{-1}(\omega T_2)$$

$$= -\tan^{-1}\left(\sqrt{\frac{T_1}{T_2}}\right) - \tan^{-1}\left(\sqrt{\frac{T_2}{T_1}}\right)$$

$$= - \left(\tan^{-1} \left(\frac{\sqrt{\frac{T_1}{T_2}} + \sqrt{\frac{T_2}{T_1}}}{1 - \sqrt{\frac{T_1}{T_2}} \sqrt{\frac{T_2}{T_1}}} \right) \right)$$

$$= - \left(\tan^{-1} \left(\frac{\sqrt{\frac{T_1}{T_2}} + \sqrt{\frac{T_2}{T_1}}}{1 - 1} \right) \right)$$

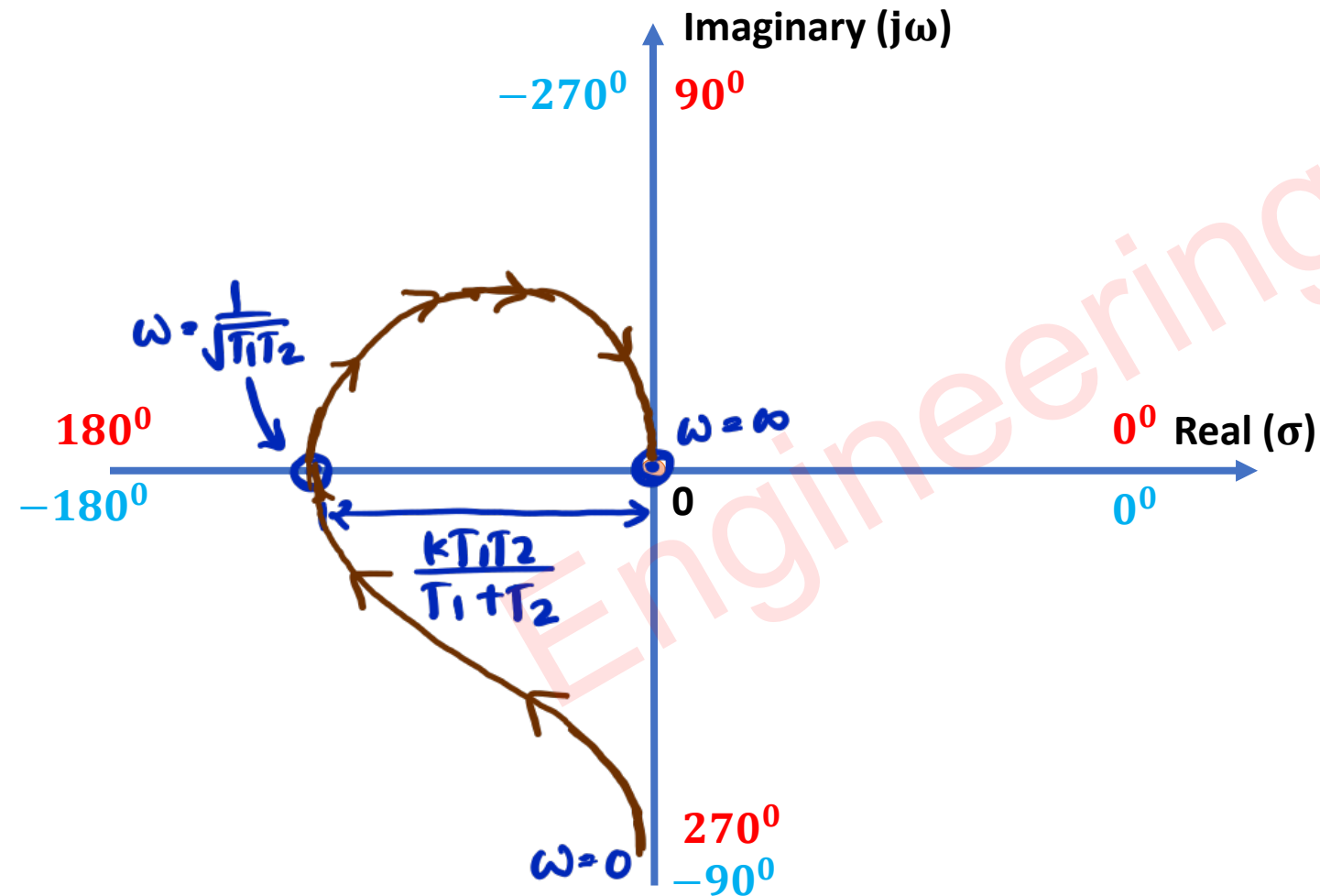
$$= -90^\circ$$

Engineering Funda

Polar Plot For Type '1' System

Example 2: Draw Polar Plot for given Open Loop Transfer Function

$$\Rightarrow G(S) = \frac{K}{S(1 + ST_1)(1 + ST_2)}$$



Step 1 - Put $s = j\omega$ in $G(s)$

$$\Rightarrow G(j\omega) = \frac{K}{j\omega(1+j\omega T_1)(1+j\omega T_2)}$$

Step 2 - Write $G(j\omega)$ in polar Form

$$\Rightarrow G(j\omega) = |G(j\omega)| \angle G(j\omega)$$

$$\Rightarrow \text{magnitude} = |G(j\omega)| = \frac{K}{\omega \sqrt{1+\omega^2 T_1^2} \sqrt{1+\omega^2 T_2^2}}$$

$$\begin{aligned} \Rightarrow \text{phase} = \angle G(j\omega) &= -\tan^{-1}\left(\frac{\omega}{0}\right) - \tan^{-1}\left(\frac{\omega T_1}{1}\right) - \tan^{-1}\left(\frac{\omega T_2}{1}\right) \\ &= -90 - \tan^{-1}(\omega T_1) - \tan^{-1}(\omega T_2) \end{aligned}$$

Step 3 - Find magnitude and phase at $\omega = 0$ and $\omega = \infty$

- For $\omega = 0$, magnitude = ∞ , phase = -90
- For $\omega = \infty$, magnitude = 0 , phase = -270

Step 4 - Separate Real and Imag. parts of $G(j\omega)$.

$$\Rightarrow G(j\omega) = \frac{K}{j\omega(1+j\omega T_1)(1+j\omega T_2)} \times \frac{(1-j\omega T_1)(1-j\omega T_2)}{(1-j\omega T_1)(1-j\omega T_2)}$$

$$\Rightarrow G(j\omega) = \frac{K(-j)(1-\omega^2 T_1 T_2 - j\omega T_1 - j\omega T_2)}{\omega(1+\omega^2 T_1^2)(1+\omega^2 T_2^2)}$$

$$\Rightarrow G(j\omega) = \underbrace{\frac{-K(\omega(T_1+T_2))}{\omega(1+\omega^2 T_1^2)(1+\omega^2 T_2^2)}}_{\text{Real } [G(j\omega)]} + j \underbrace{\frac{K(\omega^2 T_1 T_2 - 1)}{\omega(1+\omega^2 T_1^2)(1+\omega^2 T_2^2)}}_{\text{Imag } [G(j\omega)]}$$

Step 5 - Intersection to Real Axis $\Rightarrow \text{Imag. } [G(j\omega)] = 0$

$$\Rightarrow \omega = \frac{1}{\sqrt{T_1 T_2}}$$

$$\Rightarrow \text{magnitude} = |G(j\omega)| = \frac{K}{\omega \sqrt{1+\omega^2 T_1^2} \sqrt{1+\omega^2 T_2^2}}$$

$$= \frac{K}{\frac{1}{\sqrt{T_1 T_2}} \sqrt{1 + \frac{T_1}{T_2}} \sqrt{1 + \frac{T_2}{T_1}}}$$

$$= \frac{K T_1 T_2}{T_1 + T_2}$$

$$\Rightarrow \text{phase} = \angle G(j\omega) = -90 - \tan^{-1}(\omega T_1) - \tan^{-1}(\omega T_2)$$

$$= -90 - \tan^{-1}\left(\sqrt{\frac{T_1}{T_2}}\right) - \tan^{-1}\left(\sqrt{\frac{T_2}{T_1}}\right)$$

$$= -90 - \tan^{-1}\left(\frac{\sqrt{T_1/T_2} + \sqrt{T_2/T_1}}{1 - \sqrt{\frac{T_1}{T_2}} \sqrt{\frac{T_2}{T_1}}}\right)$$

$$= -90 - 90$$

$$= -180^\circ$$

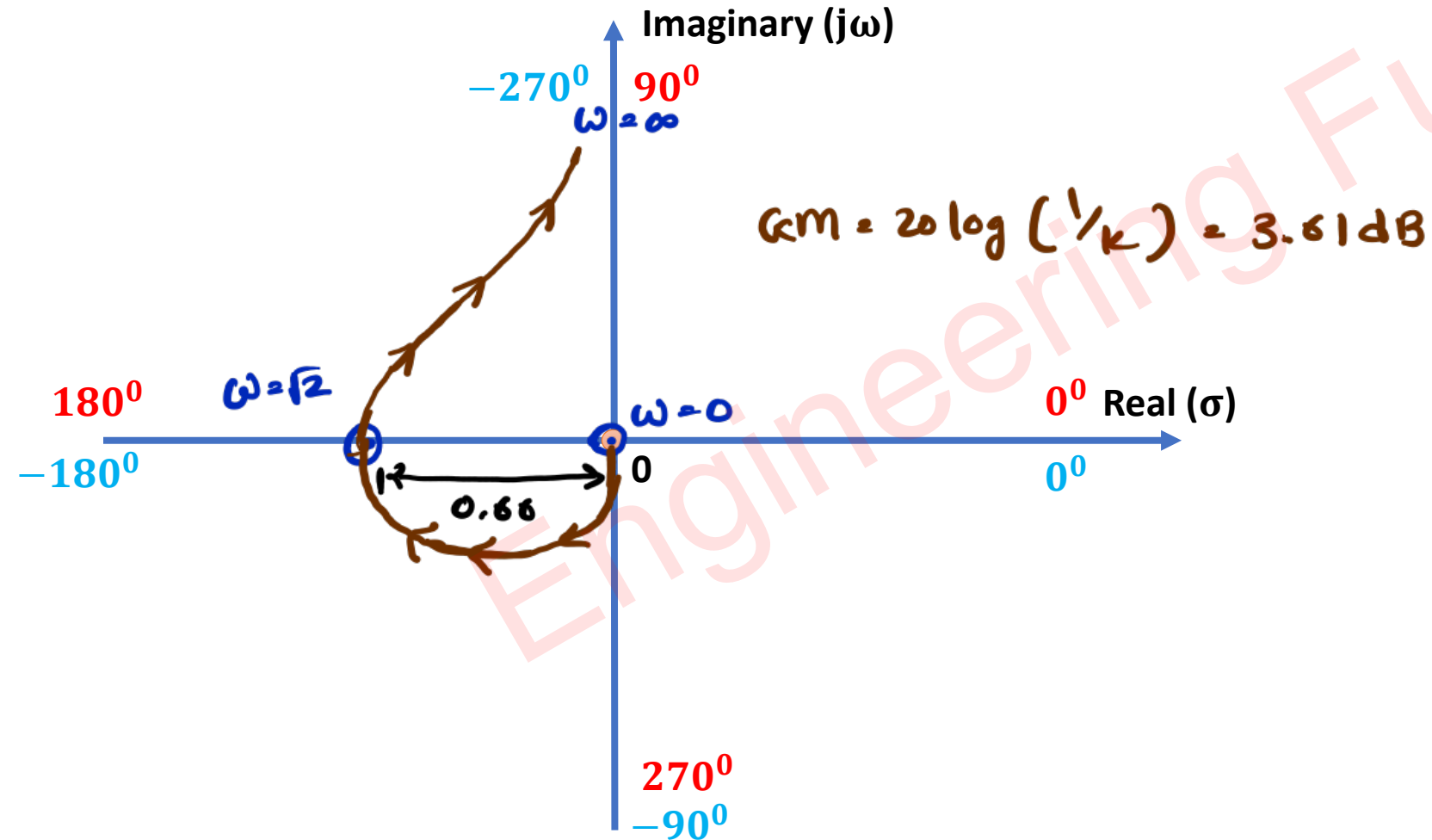
Step-6 - Intersection to Imag. Axis, $\Rightarrow \text{Real}[G(j\omega)] = 0$

$$- \omega = 0$$

Polar Plot Example

Example 3: Draw Polar Plot for given Open Loop Transfer Function

$$\Rightarrow G(S) = \frac{S^3}{(S+1)(S+2)}$$



Step-1 - Put $s = j\omega$ in $G(s)$

$$\Rightarrow G(j\omega) = \frac{(j\omega)^3}{(1+j\omega)(2+j\omega)}$$

Step-2 - Write $G(j\omega)$ polar form

$$\Rightarrow G(j\omega) = |G(j\omega)| \angle G(j\omega)$$

$$\Rightarrow |G(j\omega)| = \frac{\omega^3}{\sqrt{1+\omega^2} \sqrt{4+\omega^2}}$$

$$\Rightarrow \angle G(j\omega) = 270^\circ - \tan^{-1}(\omega) - \tan^{-1}(\omega/2)$$

Step-3 - Find magnitude and phase at $\omega = 0$ & $\omega = \infty$

- $\omega = 0$, mag. = 0, phase = 270°

- $\omega = \infty$, mag. = ∞ , phase = 90°

Step 4 - Separate Real & Imag. parts of $G(j\omega)$

$$\Rightarrow G(j\omega) = \frac{(j\omega)^3}{(1+j\omega)(2+j\omega)} \times \frac{(1-j\omega)(2-j\omega)}{(1-j\omega)(2-j\omega)}$$

$$\Rightarrow G(j\omega) = \frac{-j\omega^3(2-\omega^2-3j\omega)}{(1+\omega^2)(4+\omega^2)}$$

$$\Rightarrow G(j\omega) = \frac{-3\omega^4}{(1+\omega^2)(4+\omega^2)} + j \frac{\omega^3(\omega^2-2)}{(1+\omega^2)(4+\omega^2)}$$

$\text{Real}[G(j\omega)] \qquad \text{Imag}[G(j\omega)]$

Step 5 - Intersection to Real Axis, $\text{Imag}[G(j\omega)] = 0$

- $\omega = 0$

- $\omega = \sqrt{2}$, $\text{Mag.} = \frac{\omega^3}{\sqrt{1+\omega^2}\sqrt{4+\omega^2}} = 0.66$, $\text{phase} = 270 - \tan^{-1}(\omega) - \tan^{-1}(\omega/2)$
 $= 180$

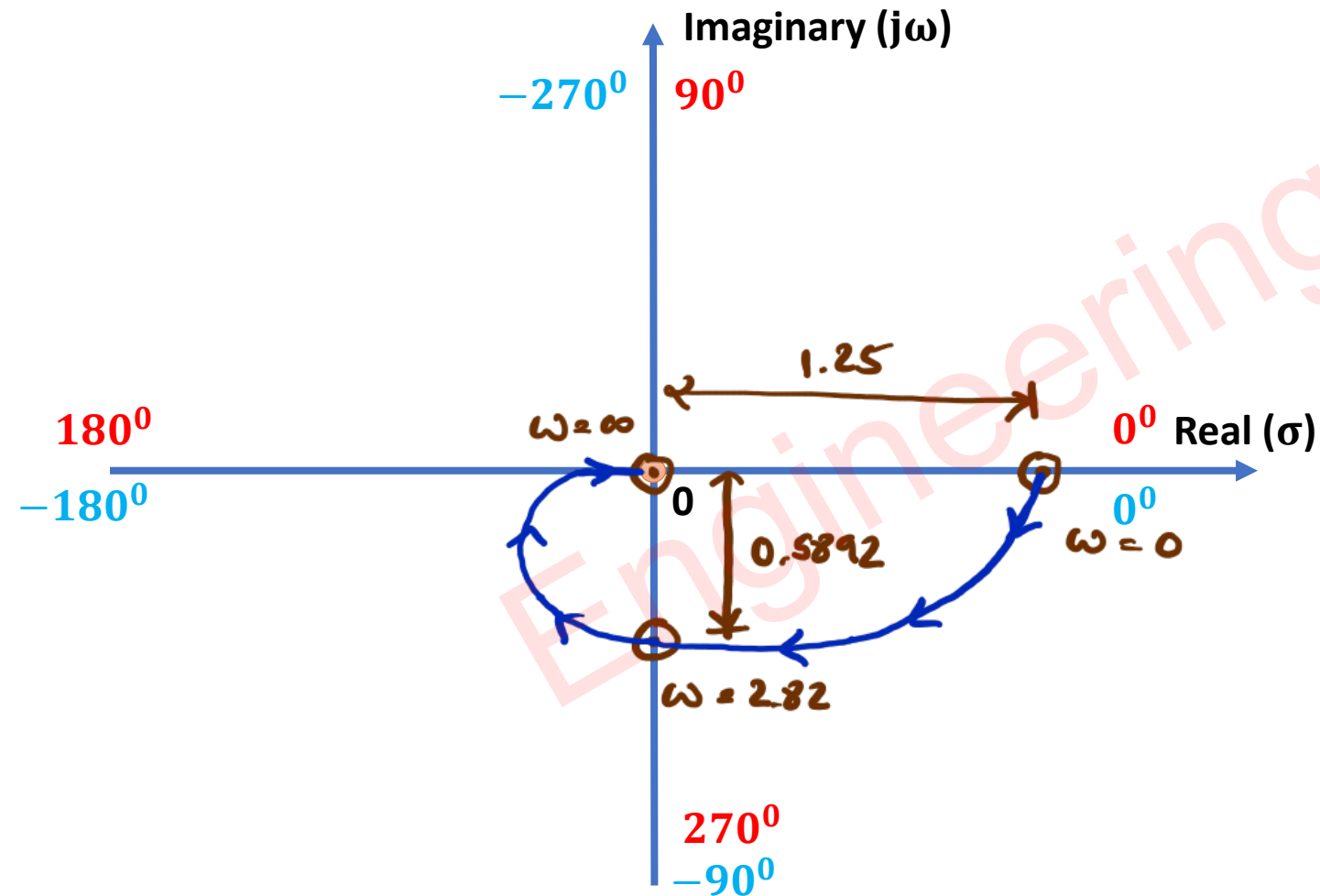
Step 6 - Intersection to Imag. Axis, $\text{Real}[G(j\omega)] = 0$

- $\omega = 0$

Polar Plot Example

Example 4: Draw Polar Plot for given Open Loop Transfer Function

$$\Rightarrow G(S) = \frac{10}{(S + 2)(S + 4)}$$



Step-1 - Put $s=j\omega$ in $G(s)$

$$\Rightarrow G(j\omega) = \frac{10}{(2+j\omega)(4+j\omega)}$$

Step-2 - Write $G(j\omega)$ in polar Form

$$\Rightarrow G(j\omega) = |G(j\omega)| \angle G(j\omega)$$

$$\Rightarrow \text{Magnitude} = |G(j\omega)| = \frac{10}{\sqrt{4+\omega^2} \sqrt{16+\omega^2}}$$

$$\Rightarrow \text{phase} = \angle G(j\omega) = -\tan^{-1}(\omega/2) - \tan^{-1}(\omega/4)$$

Step-3 - Find magnitude & phase for $\omega=0$ & $\omega=\infty$

$$- \omega=0, \text{ Mag.} = 1.25, \text{ phase} = 0$$

$$- \omega=\infty, \text{ Mag.} = 0, \text{ phase} = -180^\circ$$

Step-4 - Separate Real & Imag. parts of $G(j\omega)$

$$\Rightarrow G(j\omega) = \frac{10}{(2+j\omega)(4+j\omega)} \times \frac{(2-j\omega)(4-j\omega)}{(2-j\omega)(4-j\omega)}$$

$$\Rightarrow G(j\omega) = \underbrace{\frac{10(8-\omega^2)}{(4+\omega^2)(16+\omega^2)}}_{\text{Real } [G(j\omega)]} - j \underbrace{\frac{60\omega}{(4+\omega^2)(16+\omega^2)}}_{\text{Imag. } [G(j\omega)]}$$

Step 5 - Intersection to Real Axis, $\Rightarrow \text{Imag. } [G(j\omega)] = 0$

- $\omega = 0$, $\text{mag.} = 1.25$, $\text{phase} = 0$

Step 6 - Intersection to Imag. Axis, $\Rightarrow \text{Real } [G(j\omega)] = 0$

- $\omega = \sqrt{8} = 2.82 \text{ rad/sec}$

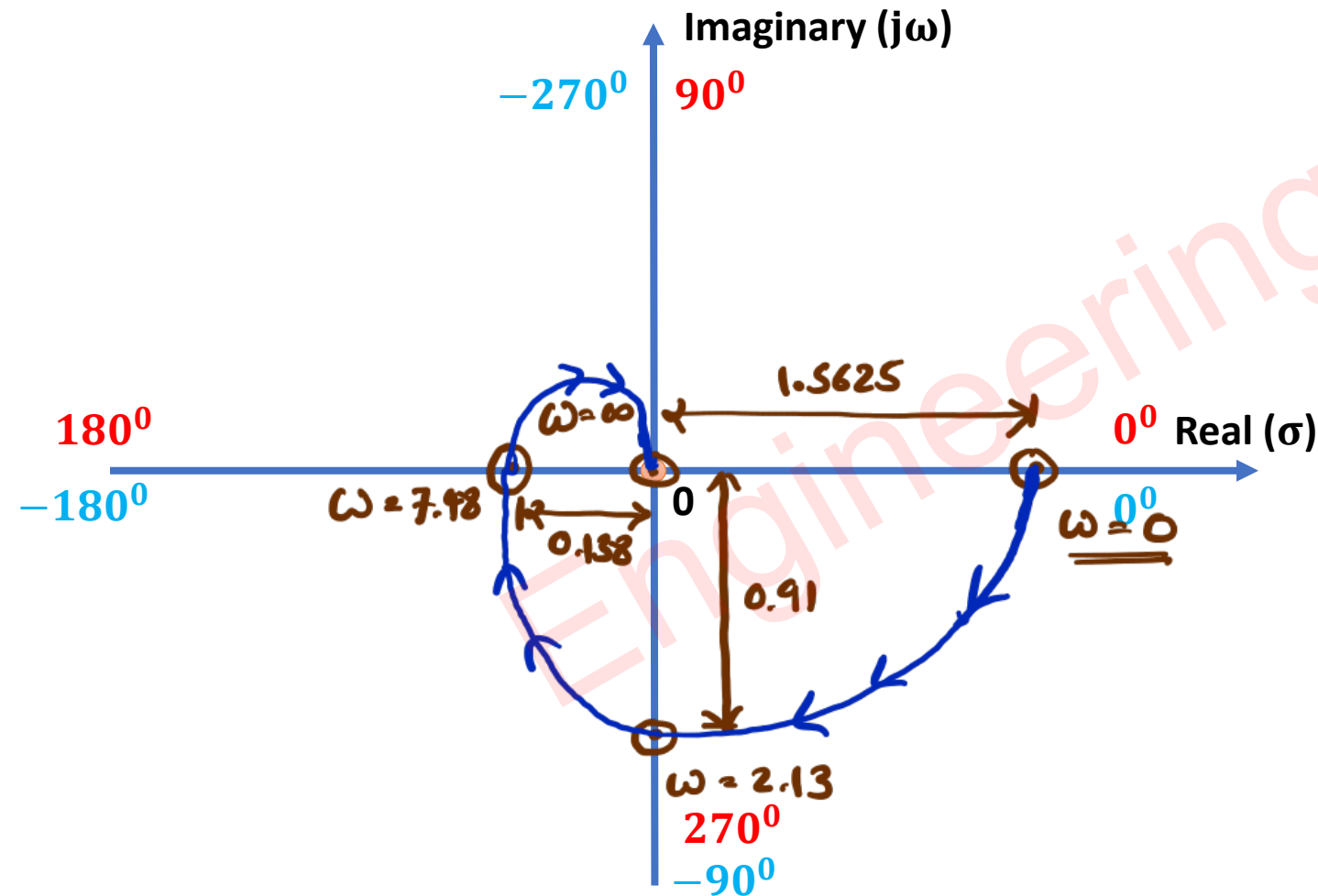
- $\text{mag.} = \frac{10}{\sqrt{4+\omega^2} \sqrt{16+\omega^2}} = 0.5892$

- $\text{phase} = -\tan^{-1}(\omega/2) - \tan^{-1}(\omega/4) = -90$

Polar Plot Example

Example 5: Draw Polar Plot for given Open Loop Transfer Function

$$\Rightarrow G(S) = \frac{100}{(S + 2)(S + 4)(S + 8)}$$



Step-1 - Put $s = j\omega$ in $G(s)$

$$\Rightarrow G(j\omega) = \frac{100}{(2+j\omega)(4+j\omega)(8+j\omega)}$$

Step-2 - Write $G(j\omega)$ in polar Form

$$\Rightarrow G(j\omega) = |G(j\omega)| \angle G(j\omega)$$

$$\Rightarrow \text{Magnitude} = |G(j\omega)| = \frac{100}{\sqrt{4+\omega^2} \sqrt{16+\omega^2} \sqrt{64+\omega^2}}$$

$$\Rightarrow \text{Phase} = \angle G(j\omega) = -\tan^{-1}(\omega/2) - \tan^{-1}(\omega/4) - \tan^{-1}(\omega/8)$$

Step-3 - Find Magnitude and phase of $\omega = 0$ & $\omega = \infty$

- at $\omega = 0$, $\text{Mag.} = 1.5625$, $\text{phase} = 0$

- at $\omega = \infty$, $\text{Mag.} = 0$, $\text{phase} = -270^\circ$

Step-4 - Separate Real & Imag. parts of $G(j\omega)$

$$\Rightarrow G(j\omega) = \frac{100}{(2+j\omega)(4+j\omega)(8+j\omega)} \times \frac{(2-j\omega)(4-j\omega)(8-j\omega)}{(2-j\omega)(4-j\omega)(8-j\omega)}$$

$$\Rightarrow G(j\omega) = \frac{100(8 - \omega^2 - j6\omega)(8-j\omega)}{(4+\omega^2)(16+\omega^2)(64+\omega^2)}$$

$$\Rightarrow G(j\omega) = \frac{100(64 - 8\omega^2 - j48\omega - j8\omega + j\omega^3 - 6\omega^2)}{(4+\omega^2)(16+\omega^2)(64+\omega^2)}$$

$$\Rightarrow G(j\omega) = \underbrace{\frac{100(64 - 14\omega^2)}{(4+\omega^2)(16+\omega^2)(64+\omega^2)}}_{\text{Real } G(j\omega)} + j \underbrace{\frac{100(-56\omega + \omega^3)}{(4+\omega^2)(16+\omega^2)(64+\omega^2)}}_{\text{Imag } G(j\omega)}$$

Steps - Intersection to Real Axis, $\text{Imag } G(j\omega) = 0$

- $\omega = 0$, $\omega = \sqrt{56} = 7.48 \text{ rad/sec}$

mag. = 0.138

phase = -180°

Step 6 - Intersection to Imag. Axis, Real $G(j\omega) = 0$

- $\omega = 2.13 \text{ rad/sec}$

mag. = 0.91

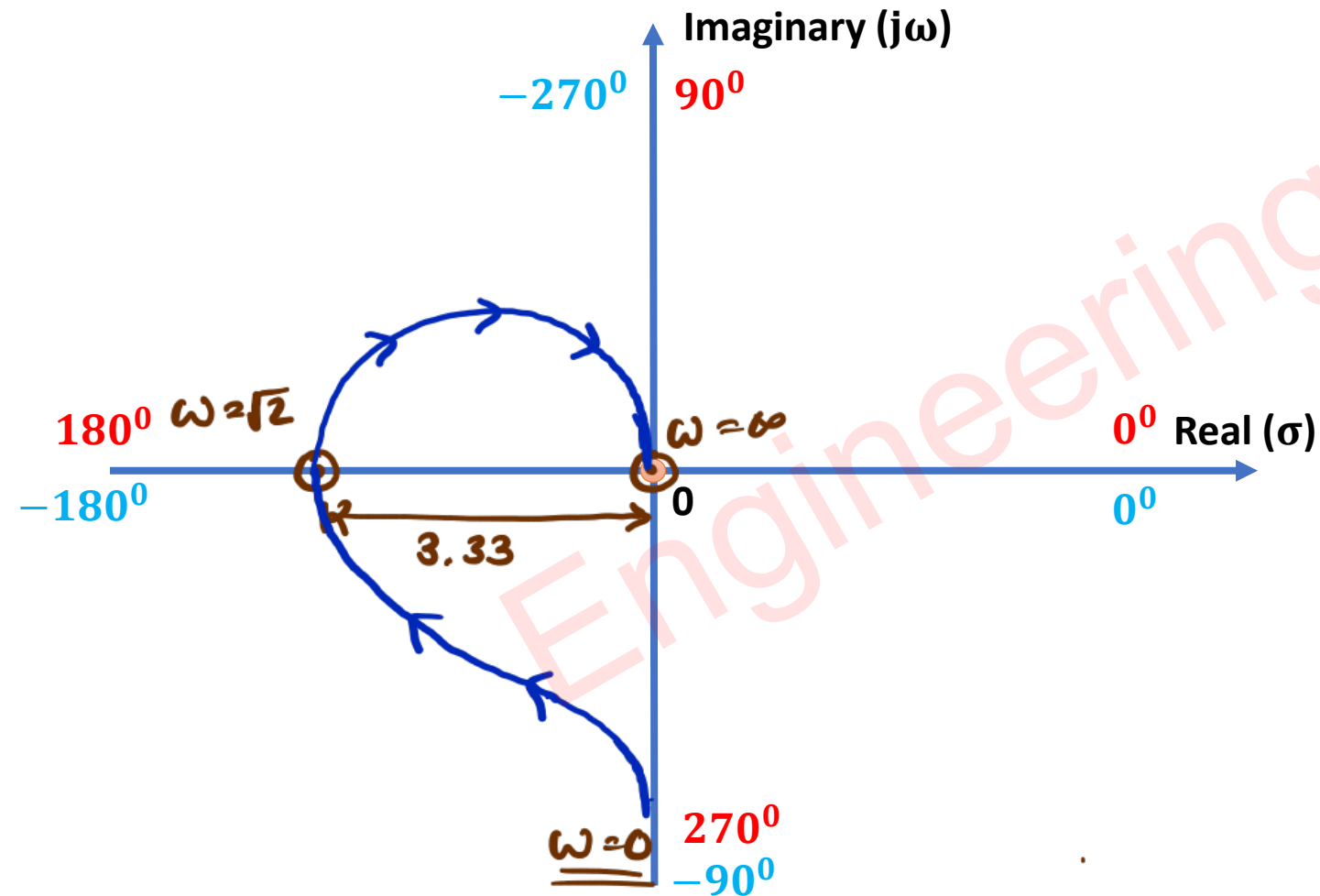
phase = -90°

Engineering Funda

Polar Plot Example

Example 6: Draw Polar Plot for given Open Loop Transfer Function

$$\Rightarrow G(S) = \frac{20}{S(S+1)(S+2)}$$



Step-1 - Put $s = j\omega$ in $G(s)$

$$\Rightarrow G(j\omega) = \frac{20}{j\omega(1+j\omega)(2+j\omega)}$$

Step-2 - Write $G(j\omega)$ in Polar Form

$$\Rightarrow G(j\omega) = |G(j\omega)| \angle G(j\omega)$$

$$\Rightarrow \text{Magnitude} = |G(j\omega)| = \frac{20}{\omega \sqrt{1+\omega^2} \sqrt{4+\omega^2}}$$

$$\Rightarrow \text{Phase} = \angle G(j\omega) = -90 - \tan^{-1}(\omega) - \tan^{-1}(\omega/2)$$

Step-3 - Find Magnitude & phase at $\omega=0$ & $\omega=\infty$

- at $\omega=0$, Mag. = ∞ , phase = -90

- at $\omega=\infty$, Mag. = 0 , phase = -270

Step-4 - Separate Real & Imag. parts of $G(j\omega)$

$$\Rightarrow G(j\omega) = \frac{20}{j\omega(1+j\omega)(2+j\omega)} \times \frac{(1-j\omega)(2-j\omega)}{(1-j\omega)(2-j\omega)}$$

$$\Rightarrow G(j\omega) = \frac{-j20(2-\omega^2-3j\omega)}{\omega(1+\omega^2)(4+\omega^2)}$$

$$\Rightarrow G(j\omega) = \underbrace{\frac{-60\omega}{\omega(1+\omega^2)(4+\omega^2)}}_{\text{Real } G(j\omega)} + j \underbrace{\frac{20(\omega^2-2)}{\omega(1+\omega^2)(4+\omega^2)}}_{\text{Imag } G(j\omega)}$$

Step-5 - Intersection to Real Axis, $\text{Imag}(G(j\omega)) = 0$

$$- \omega = \sqrt{2}, \text{ mag} = 3.33, \text{ phase} = -180^\circ$$

Step-6 - Intersection to Imag Axis, $\text{Real}(G(j\omega)) = 0$

$$- \omega = 0$$